NATHANAEL OH

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EDUCATION

BS Johns Hopkins University, Computer Science — May 2025

- GPA: 3.90
- Honors:
  - Dean's List ('21-'22, '22-'23)
- Relevant coursework:
  - Gateway Java
  - Intermediate Programming (C/C++)
  - Fullstack JavaScript
  - Data Structures
  - Intro to Algorithms

HSD McLean High School, Advanced Diploma — May 2021

- GPA: 4.3
- Honors:
  - Scholar-Athlete Award
  - AP Scholar

SKILLS

Java

Extensive proficiency in Java, encompassing advanced concepts such as Abstract classes, Interfaces, and Generics. Applied expertise in JUnit for comprehensive testing. Demonstrated ability through the development of diverse projects including GUIs, card games, and data analysis applications. School project repository is currently private, but can be made accessible upon request: https://github.com/bathan1/gateway-java-projects

C/C++

Possess a solid grasp of C/C++, including manual memory management, object-oriented programming principles, and proficient debugging skills using gdb and valgrind. Applied this knowledge in various projects encompassing maze pathing, data storage, linked lists, trie data structures, and .ppm image handling. The school project repository is presently private but can be shared upon request: https://github.com/bathan1/intermediate-programming

Web Development

Demonstrated expertise in deploying full-stack applications, employing a skill set that encompasses HTML, CSS (including Tailwind CSS), and React for robust client-side development. Leveraging ExpressJS and NodeJS, I proficiently construct custom RESTful APIs, seamlessly bridging the gap between front-end and server, and ensuring smooth interactions with MongoDB databases. Additionally, I am adept at utilizing advanced development tools such as Vite, along with npm and pnpm package managers for efficient project management.

Python

Proficient in the Python language with a strong understanding of the pip/venv model for managing Python modules. Experienced in utilizing data science packages such as Pandas to efficiently clean and prepare data for handling at Children's National Hospital.

Git/GitHub version control

Accumulated one year of hands-on experience in employing Git and GitHub for version control. Proficient in collaborative workflows, utilizing Git/GitHub to manage and track code changes effectively.

PROFESSIONAL EXPERIENCE

Paper Tutor, Online 2022 to Present

Math/Science Tutor

- Tutor math and science online to numerous K-12 students simultaneously in a fast-paced classroom setting.

Children's National Hospital, Washington D.C. 2022

Project Intern / Research Assistant for Dr. Natasha Shur

- Team member for Rhabdomyolysis project.
- Researched the use of dexamethasone in treating Rhabdomyolysis.
- Submitted case report on Waardenburg Syndrome to Clinical Pediatrics

Johns Hopkins University, Baltimore MD 2022-2023

Data entry / Research Assistant for Dr. Richard Skolasky.

- Conducted 500+ survey calls to spine surgery outpatients to record outpatient data.
- Entered and edited over 100 patient data on the REDCap survey platform.

Children's National Hospital, Washington D.C. 2019 to present

Research Coordinator for Dr. Gary Rogers

- Managed over 5 medical databases used for academic publications.
- Handled IRB related processes and paper for team.
- Retrospective diagnosis of normocephalic sagittal craniosynostosis in young children via CT scan analysis.
- Experience using Slicer3D to recreate 3D images of CT scans.
- Analysis of facial nerve activity in infant patients undergoing Mandibular Distraction Osteogenesis.
- Assisted with the writeup of 2 academic papers (see journal publications).

PROJECTS

Web apps:

Built using a variety of technologies such as ReactJS, MongoDB, NodeJS, and ExpressJS.

- Bio-Match: A web app designed to connect users to clinical trials listed based on a sorting algorithm that sorts clinical trials based on the user's profile. Built for the HopHacks 2023 Hackathon along with 3 other collaborators.
    - GitHub repo: https://github.com/xerotresdin/BioMatch
    - Render deployment: https://bio-match.onrender.com/

PUBLICATIONS

Journal Publications

Crowder, H.R; Oh, N.S; Mantilla-Rivas, E; Manrique, M; Rogers, G.F; Oh, A.K. Intra-op nerve monitoring in patients with Robin Sequence undergoing Mandibular Distraction Osteogenesis.

McGrath, J.L; Manrique, M; Mantilla-Rivas, E; Rana, Md.S Crowder, H.R; Ichiuji, B.A; Aivaz, M; Oh, N.S; Oh, A.K; Rogers, G.F; The Prevalence of Normocephalic Sagittal Craniosynostosis in Young Children

Journal Papers in Review

Oh, Nathanael Suhyun, Shur, Natasha., and Hain, Heather, "A Waardenburg case and genetic testing complexity in diverse populations," Submitted to: Clinical Pediatrics.

REFERENCES

Available upon request.